

- iii) $f(x) = \frac{x^2-4}{x-2}$. What should be the value of $f(2)$, so that $f(x)$ is continuous at $x=2$?

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9.7 LET US SUM UP

In this unit we have discussed the basic concepts of limit and continuity which form the basis for the rest of the differential calculus. In this respect, first, we discussed different types (Left-hand, right-hand etc.) of limit of a function, and questions and conditions for its existence. Also, we discussed algebra of limits, and related theorems. Using the concept of limit, we defined and discussed the concepts of *local* continuity and discontinuity, and related conditions for a function to be continuous at a point. Finally, some properties of continuous functions are discussed.

9.8 KEY WORDS

- Limit** : Limit is a basic concept in calculus and mathematical analysis, for determining the behaviour of a function around a point/ value in the domain of the function. It forms the basis of the more advanced concepts, in calculus, of continuity and differentiability of functions.
- Continuity** : Graphically, a function $f(x)$ is continuous at a point $x = a$ provided its graph in the surrounding of the point a can be drawn without lifting the pen/ pencil. Formally, a function $f(x)$ is said to be continuous for $x = a$, provided $\lim_{x \rightarrow a} f(x)$ exists, is finite and is equal to $f(a)$.

9.9 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

$$\begin{aligned}
 1) \quad i) \quad & \lim_{x \rightarrow 0} \frac{\sqrt{a+x} - \sqrt{a-x}}{x} \\
 &= \lim_{x \rightarrow 0} \frac{a+x - a-x}{x(\sqrt{a+x} - \sqrt{a-x})} \\
 &= \lim_{x \rightarrow 0} \frac{2x}{x(\sqrt{a+x} - \sqrt{a-x})} = \lim_{x \rightarrow 0} \frac{2}{\sqrt{a+x} + \sqrt{a-x}}
 \end{aligned}$$

(Since $x \rightarrow 0$ and $x \neq 0$ we can cancel x in the ratio)

Now, as $x \rightarrow 0$, $\sqrt{a \pm x} \rightarrow \sqrt{a}$. Thus, the limit is $\frac{2}{2\sqrt{a}} = \frac{1}{\sqrt{a}}$.

ii) $|\sin x - 0| = |\sin x|$ can be made arbitrarily small by making $|x|$ arbitrarily small.

Thus, $\lim_{x \rightarrow 0} \sin x = 0$

iii) $\sin x - \sin \theta = 2 \sin \frac{1}{2}(x - \theta) \cos \frac{1}{2}(x + \theta)$

As, $x \rightarrow 0$, $\sin \frac{1}{2}(x - \theta)$.

Also $\left| \cos \frac{1}{2}(x + \theta) \right| \leq 1$.

Thus, $\lim_{x \rightarrow 0} (\sin x - \sin \theta) = 0$, i.e., $\lim_{x \rightarrow \theta} \sin x = \sin \theta$

2) i) $\lim_{x \rightarrow 1+0} f(x) = 10$, $\lim_{x \rightarrow 1-0} f(x) = -10$

Thus, $\lim_{x \rightarrow 1} f(x) = 10$ does not exist; whereas $f(1)$ exists $= 1$.

ii) $\lim_{x \rightarrow 0+0} f(x) = 10$, $\lim_{x \rightarrow 0-0} f(x) = 1$, which is not equal to $f(0) = -1$.

3) i) Put $x^2 = z$ as $x \rightarrow 0$, $z \rightarrow 0$

$$= \lim_{z \rightarrow 0} \frac{\sin z}{z} \cdot \sqrt{z} = \left(\lim_{z \rightarrow 0} \frac{\sin z}{z} \right) \left(\lim_{z \rightarrow 0} \sqrt{z} \right) = 1.0 = 0.$$

$$\text{ii) } \lim_{x \rightarrow 0} \frac{1 - \cos x}{x \sin x} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x \cdot 2 \sin \frac{x}{2} \cos \frac{x}{2}}$$

$$\lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{x \cos \frac{x}{2}} = \frac{1}{2} \left(\lim_{\frac{x}{2} \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \right) \cdot \frac{1}{\left(\lim_{\frac{x}{2} \rightarrow 0} \cos \frac{x}{2} \right)} = \frac{1}{2} \cdot 1 \cdot \frac{1}{1} = \frac{1}{2}$$

iii) Put $\frac{1}{x} = z$ as $x \rightarrow \infty$, $z \rightarrow 0$, $\lim_{z \rightarrow 0} z \sin \frac{1}{z}$. Since z is not actually zero, value of $\sin \frac{1}{z}$ is a finite quantity between $[-1, 1]$. Thus, by taking z as small as we like we can make $z \sin \frac{1}{z}$ as small as we like, i.e., $\left| z \sin \frac{1}{z} - 0 \right|$ is less than any pre-assigned positive quantity. Thus, the limit is zero.

$$\text{iv) } \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} =$$

$$\lim_{h \rightarrow 0} \frac{a(x+h)^2 + b(x+h) + c - ax^2 - bx - c}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a(x^2 + 2xh + h^2) + bh - ax^2}{h} = \lim_{h \rightarrow 0} \left(\frac{2axh + h^2 + bh}{h} \right)$$

$$= \lim_{n \rightarrow 0} (2ax + b) = 2ax + b \quad [\because h \neq 0]$$

$$v) \lim_{n \rightarrow \infty} \frac{n(n+1)(2n+1)}{6n^3} = \lim_{n \rightarrow \infty} \frac{1}{6} \left(1 + \frac{1}{n}\right) \left(2 + \frac{1}{n}\right) = \frac{1}{6} \cdot 1 \cdot 2 = \frac{1}{3}$$

vi) Put $\tan^{-1} x = \theta$, i.e., $\tan \theta = x$. As $x \rightarrow 0$, $\theta \rightarrow 0$. Now,

$$\lim_{\theta \rightarrow 0} \frac{\theta}{\tan \theta} = \left(\lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} \right) \cdot \left(\lim_{\theta \rightarrow 0} \cos \theta \right) = 1 \cdot 1 = 1.$$

Check Your Progress 2

1) i) $\lim_{x \rightarrow 0} f(x) = 0$ (It has already been proved in the problem for limit).

Thus,

$$\lim_{x \rightarrow 0} f(x) = f(0) \text{ and } f(x) \text{ is continuous at } x = 0.$$

ii) $f(x) = \frac{x^2 - 2x + 4}{x^2 - 5x + 6}$ is the ratio of two continuous functions.

(Polynomials are continuous can be verified easily). Thereby the property – III of the continuous function, $f(x)$ will be continuous at all values of x except when $x^2 - 5x + 6$ equals zero, i.e., the point of discontinuity of $f(x)$ are $x = 2, 3$.

iii) $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{(x+2)(x-2)}{(x-2)} = 4$. Thus, $f(2) = 4$ is the requirement for $f(x)$ to be continuous at $x = 2$.

9.10 EXERCISES

Q1. Show $\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}$, for all natural numbers n , provided $a > 0$.

Ans. Expanding, by binomial theorem, the numerator of the expression

$$\frac{x^n - a^n}{x - a},$$

$$= (x - a)(x^{n-1} + x^{n-2} \cdot a + \dots + x \cdot a^{n-2} + a^{n-1}) / (x - a)$$

For all values of x , except for $x = a$, we have

$$= (x^{n-1} + x^{n-2} \cdot a + \dots + x \cdot a^{n-2} + a^{n-1}); \text{ for } x \neq a$$

$$= (x^{n-1} + x^{n-2} \cdot a + \dots + x \cdot a^{n-2} + a^{n-1}); \text{ for } x \neq a$$

By the definition of limit, we know for taking limit at a point, say a , the value at that point is not taken into consideration. Hence

$$\lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = \lim_{x \rightarrow a} (x^{n-1} + x^{n-2} \cdot a + \dots + x \cdot a^{n-2} + a^{n-1}); \quad \dots (i)$$

From algebra of limits, we know

$$\lim_{X \rightarrow a} X^i = (\lim_{X \rightarrow a} X)^i = a^i \text{ for } i = 1, 2, 3, \dots, \quad \dots \text{ (ii)}$$

and

$$\lim_{X \rightarrow a} k \cdot X = k \cdot (\lim_{X \rightarrow a} X), \text{ for any constant real number } k \neq 0. \quad \dots \text{ (iii)}$$

Using (i), (ii), and (iii), we get

$$\begin{aligned} \lim_{X \rightarrow a} (x^{n-1} + x^{n-2} \cdot a + \dots + x \cdot a^{n-2} + a^{n-1}) \\ = (a^{n-1} + a^{n-2} \cdot a + \dots + a \cdot a^{n-2} + a^{n-1}) = n a^{n-1}. \end{aligned}$$

Q2. Show $\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = n$

Ans. For

$$\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = n$$

As above, expanding the numerator, by binomial theorem

$$\frac{(1+x)^n - 1}{x}$$

we get

$$\begin{aligned} &= \{(1 + n x + n(n-1)/2 x^2 + \dots + n x^{n-1} + x^n) - 1\} / x \\ &= n x \cdot (1 + \{(n-1)/2\} x + \dots + x^{n-1}) / x \end{aligned}$$

As we take limit at $x=0$, the value at which is not taken into consideration, hence, cancelling out x from numerator and denominator, we get

$$\lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = \lim_{x \rightarrow 0} n (1 + \{(n-1)/2\} x + \dots + x^{n-1}) = n$$

Q3. Find $\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$

Ans. For

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0$$

From Archimedean property of real numbers, we know that given real numbers $x > 0$ and $y=1$, there exists natural number k , such that $x < k \cdot 1 < m \cdot 1$, for all $m > k$.

In other words, there exists a natural number k , such that

$$(x/n) < 1 \text{ for all } n \geq m > k \quad \dots \dots \dots \text{ (i)}$$

Consider $\frac{x^n}{n!} = \left[\frac{x^k}{k!} \right] \frac{x}{k+1} \frac{x}{k+2} \dots \frac{x}{n} \quad \dots \text{ (ii)}$

Let $\frac{x}{k+1} = y$ with $y < 1 \dots \dots$ (iii)

and $\frac{x}{j} < \frac{x}{k+1} = y < 1$ for all $j > k+1$,

By using (ii) in (iii), we get

$$\frac{x^n}{n!} < \left[\frac{x^k}{k!} \right] \cdot y \cdot y \cdot y = \left[\frac{x^k}{k!} \right] \cdot (y)^{n-k} \dots \dots \text{(iv)}$$

Now $\left[\frac{x^k}{k!} \right] (1/y)^k$ is a constant, say C .

Therefore, (iv) implies

$$\frac{x^n}{n!} < C (y)^n, \text{ where } y < 1 \dots \dots \text{(v)}$$

R.H.S is a geometric series with ratio $y < 1$. Hence, it converges to 0.

Then (v) implies $\frac{x^n}{n!}$ converges to 0, or

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0.$$

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UNIT 10 DIFFERENTIAL CALCULUS: FUNCTIONS OF ONE VARIABLE

Structure

- 10.0 Objectives
- 10.1 Introduction
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- 10.3 Geometrical Interpretation of Derivatives
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10.0 OBJECTIVES

After going through this unit, you should be able to

- explain the concept of derivatives;
- present geometric interpretation of derivatives;
- solve the applications of simple derivatives;
- identify the slope and curvature of a curve; and
- explain the mean value theorem and L'Hopital's rule.

10.1 INTRODUCTION

In this unit, we will understand the concept of change with the help of derivatives. To understand change, it is imperative that we spend some time on understanding the geometric interpretations of derivatives. Once we grasp this part, we will be in a position to understand the economic applications of the simple derivatives. You will be given an exposure of some solved applications which are common in economics. Further, we will also learn about slope and curvature and how are they related. Finally, we will finish the unit with Taylor series, mean value theorem and L'Hopital's rule.

10.2 DERIVATIVE

The **derivative** of a function measures the function's rate of change. To understand the concept, consider the **increment** of a variable (x) in changing from one value to another is the **difference** obtained by subtracting the first value from the second. An **increment** of x is denoted by Δx (delta x) or h .

If in a function $y = f(x)$, the independent variable x takes an increment Δx (or h), then Δy (or k) denotes the corresponding increment of y , i.e., of $f(x)$.

Thus, we have, $y + \Delta y = f(x + \Delta x)$ i.e., $\Delta y = f(x + \Delta x) - f(x)$

or, $y + k = f(x + h)$ i.e., $k = f(x + h) - f(x)$

Example 10.1: Let $y = +\sqrt{x}$

Suppose, x increases from 4 to 4.1, i.e., $\Delta x = 0.1$, then y increases from 2 to 2.025 (approx), i.e., $\Delta y = 0.025$. Suppose x decreases from 4 to 3.9, i.e., $\Delta x = -0.1$, then y decreases from 2 to 1.975, i.e., $\Delta y = -0.025$.

Increments are always reckoned from the arbitrarily fixed initial value of the independent variable x .

From a fixed initial value 4 of x , if x increases successively to 4.1, 4.01, 4.001, etc., then although the corresponding increments $\Delta x = (0.1, 0.01, 0.001, \dots)$ and $\Delta y = (0.025, 0.01, 0.001, \dots)$ are getting smaller, their ratio $\frac{\Delta y}{\Delta x}$ being 0.025, 0.01, 0.001, ... (approx.) is approaching a definite number 0.25. Thus, the ratio of the increments can be brought as near to 0.25 as we want by making Δx approaching zero.

Let $y = f(x)$ be a finite function defined in any interval of x and assume x to have any particular value in the interval. Let Δx (or h) be the increment of x , and let Δy (or k) $= f(x + \Delta x) - f(x)$ be the corresponding increment of y . If the ratio $\frac{\Delta y}{\Delta x}$ of these increments tends to a definite finite limit as Δx tends to zero, then this limit is called the **differential coefficient** (or **derivative**) of $f(x)$ (or, y) for the particular value of x and is denoted by $f'(x)$, $\frac{d}{dx}\{f(x)\}$, $\frac{dy}{dx}$ or

$$D\{f(x)\} \text{ i.e., } \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \lim_{x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \lim_{x \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

(provided this limit exists).

If as $\Delta x \rightarrow 0$, $\frac{\Delta y}{\Delta x} \rightarrow \pm \infty$, we also say the **derivative** exists.

The process described above is called **differentiation** of the function $f(x)$ **with respect to** the independent variable x .

A function to be **differentiable** at a point must be **continuous** at that point. However, if a function is continuous at a point, it may not be differentiable at that point.

Example 10.2: $f(x) = |x|$

This function is continuous at $x = 0$. But $f'(0) = \lim_{h \rightarrow 0} \frac{f(0 + h) - f(0)}{h}$

$$= \lim_{h \rightarrow 0} \frac{f(h)}{h}$$

$$\lim_{h \rightarrow 0+0} \frac{f(h)}{h} = \lim_{h \rightarrow 0+0} \frac{h}{h} = 1 \text{ or, } \lim_{h \rightarrow 0-0} \frac{f(h)}{h} = \lim_{h \rightarrow 0-0} \frac{-h}{h} = -1$$

Thus, $f'(0)$ does not exist at $x = 0$.

A function may be discontinuous at a point, but may have infinite **derivative** at that point.

Example 10.3:

$$f(x) = \begin{cases} 0, & \text{when } 0 \leq x < \frac{1}{2} \\ 1, & \text{when } x = \frac{1}{2} \\ 2, & \text{when } \frac{1}{2} < x \leq 1 \end{cases}$$

Now, $\lim_{h \rightarrow \frac{1}{2}+0} f(x) = 2$ and $\lim_{h \rightarrow \frac{1}{2}-0} f(x) = 0$

Thus, $f(x)$ is not continuous at $x = \frac{1}{2}$. But,

$$f'\left(\frac{1}{2}\right) = \lim_{h \rightarrow 0} \frac{f\left(\frac{1}{2} + h\right) - f\left(\frac{1}{2}\right)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f\left(\frac{1}{2} + h\right) - \frac{1}{2}}{h}$$

$$\lim_{h \rightarrow 0+0} \frac{f\left(\frac{1}{2} + h\right) - \frac{1}{2}}{h} = \lim_{x \rightarrow 0+0} \frac{2 - \frac{1}{2}}{h} = \infty$$

$$\lim_{h \rightarrow 0-0} \frac{f\left(\frac{1}{2} + h\right) - \frac{1}{2}}{h} = \lim_{x \rightarrow 0-0} \frac{2 - \frac{1}{2}}{h} = \infty$$

;

Thus, $f'(1/2)$ exists and is finite.

The **right-hand limit** $\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$ for any particular value of x (when exists) is known as the **right-hand derivative** of $f(x)$ at that point and is denoted by $Rf'(x)$. Similarly, the **left-hand limit** $\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$ is known as the **left-hand derivative** of $f(x)$ at x and is denoted by $Lf'(x)$. Only when both $Rf'(x)$ and $Lf'(x)$ exist and are equal, $f'(x)$ exists at x .

Now, we illustrate by a simple example how to find the **derivative** when a particular functional form is specified.

Take $f(x) = 2x^2$ such that $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \Rightarrow$

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{2(x^2 + 2xh - h^2) - 2x^2}{h} &= 4x - 2 \lim_{h \rightarrow 0} h = 4x \\ &= \lim_{h \rightarrow 0} \frac{2h(2x - h)}{h} = \lim_{h \rightarrow 0} (2x - h) \end{aligned}$$

Below are the **derivatives** (without derivation from the 'first principle') for some standard functional specifications.

i) $\frac{d(ax^n)}{dx} = anx^{n-1}$ (for all rational n)

ii) $\frac{d(e^{mx})}{dx} = me^{mx}$ (for all rational m)

iii) $\frac{d(a^x)}{dx} = a^x \log_e a$

iv) $\frac{d}{dx}(\log x) = \frac{1}{x}$

v) $\frac{d}{dx}(\sin x) = \cos x$ (x is radian)

vi) $\frac{d}{dx}(\cos x) = -\sin x$ (x is radian)

vii) $\frac{d}{dx}(\tan x) = \sec^2 x$ ($x \neq \frac{1}{2}(2n+1)\pi$)

viii) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$ ($x \neq n\pi$)

ix) $\frac{d}{dx}(\sec x) = \sec x \tan x$ ($x \neq \frac{1}{2}(2n+1)\pi$)

x) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$ ($x \neq n\pi$)

Now, we can state some important **rules of differentiation**.

- i) $\frac{d}{dx}(c) = 0$, where c is constant.
- ii) $\frac{d}{dx}[c\phi(x)] = c \frac{d}{dx}\{\phi(x)\}$, where c is constant.
- iii) $\frac{d}{dx}[\phi(x) \pm \psi(x)]$
 $= \frac{d\phi(x)}{dx} \pm \frac{d\psi(x)}{dx}$

The above result can be extended for any finite number of functions.

iv) $\frac{d}{dx}[\phi(x) \cdot \psi(x)] = \phi'(x) \cdot \psi(x) + \phi(x) \cdot \psi'(x)$

This result can also be extended for any finite number of functions.

v) $\frac{d}{dx} \left\{ \frac{\phi(x)}{\psi(x)} \right\} = \frac{\phi(x) \cdot \psi'(x) - \psi(x) \cdot \phi'(x)}{\{\psi(x)\}^2}$

vi) Let, $y = f(v)$ and $v = \phi(x)$. $f(v)$ and $\phi(x)$ are continuous (i.e., y is a continuous function of x),

Then $\frac{dy}{dx} = \frac{dy}{dv} \frac{dv}{dx}$

This rule is called **differentiation of function of a function**.

In general, let $y = f(v)$; $v = \phi(w)$, $w = \psi(x)$, then, $\frac{dy}{dx} = \frac{dy}{dv} \frac{dv}{dw} \frac{dw}{dx}$.

An implication of the above rule is $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$, provided both the derivatives exist.

Also, let $y = \psi(t)$, $x = \phi(t)$, then, $\frac{dy}{dx} = \frac{dy}{dt} / \frac{dx}{dt} \left(\frac{dx}{dt} \neq 0 \right)$

Here, the third variable t is known as the **parameter**.

We shall illustrate the importance of the above-mentioned results with some examples.

You may try and find $\frac{dy}{dx}$ in the following cases:

i) $y = \sin 2x - 2e^x + 4^4$

ii) $y = 3x^2 \operatorname{cosec} x$

iii) $y = \frac{\log x - \sin x}{2x^2 - 1}$

iv) $y = 2\cos v + \tan^2 v$

$v = e^{3x}$

v) $y = a \cos \phi, x = b \sin \phi$